

Code:

```
#include <stdio.h>
#include <stdlib.h>

// Structure to represent a block
struct Block {
    int blockNumber;
    char data[256]; // Adjust size if needed
};

int main()
{
    FILE *file;
    struct Block block;
    int blockNumber;

    // Index block to store positions
    int indexBlock[100] = {0}; // Adjust size if needed

    // Open file in write mode
    file = fopen("indexed_file.txt", "w");
    if (file == NULL)
    {
        printf("Error opening the file.\n");
        return 1;
    }

    // Write blocks
    printf("Enter blocks (Enter '0' as block number to exit):\n");

    while (1)
    {
        printf("Block Number: ");
        scanf("%d", &block.blockNumber);

        if (block.blockNumber == 0)
        {
            break;
        }

        printf("Data: ");
        scanf("%[^\n]", block.data);

        // Write block
        fwrite(&block, sizeof(struct Block), 1, file);

        // Store position in index
        indexBlock[block.blockNumber] = ftell(file) - sizeof(struct Block);
    }

    fclose(file);

    // Open file in read mode
    file = fopen("indexed_file.txt", "r");
    if (file == NULL)
    {
        printf("Error opening the file.\n");
        return 1;
    }

    // Read specific block
    while (1)
    {
        printf("Enter the block number to read (0 to exit): ");
        scanf("%d", &blockNumber);

        if (blockNumber == 0)
        {
            break;
        }

        if (indexBlock[blockNumber] == 0)
        {
            printf("Block not found.\n");
        }
        else
        {
            // Move to block location
            fseek(file, indexBlock[blockNumber], SEEK_SET);
            fread(&block, sizeof(struct Block), 1, file);

            printf("Block Number: %d\n", block.blockNumber);
            printf("Data: %s\n", block.data);
        }
    }

    fclose(file);
    return 0;
}
```

Output:

```
C:\Users\Devi\Downloads\Unt  X + v
Enter blocks (Enter '0' as block number to exit):
Block Number: 5
Data: ASWATHA
Block Number: 1
Data: SATHYA
Block Number: 0
Enter the block number to read (0 to exit): 2
Block not found.
Enter the block number to read (0 to exit): 5
Block Number: 5
Data: ASWATHA
Enter the block number to read (0 to exit):
```